

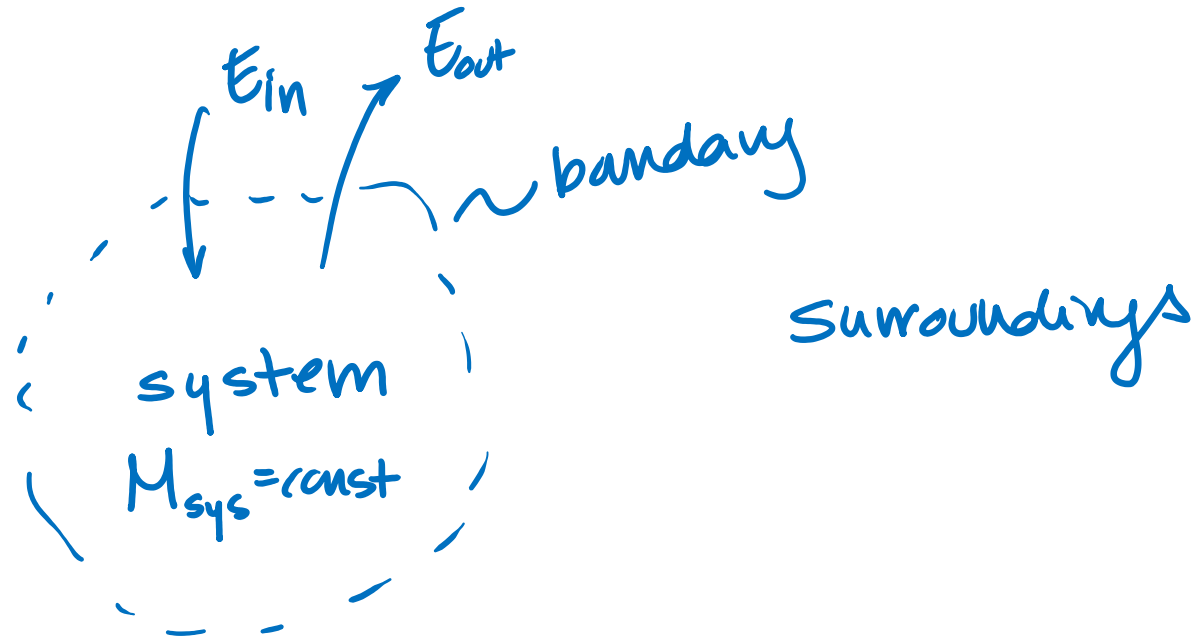


# ME 300: Engineering Thermodynamics

## Systems and Control Volumes

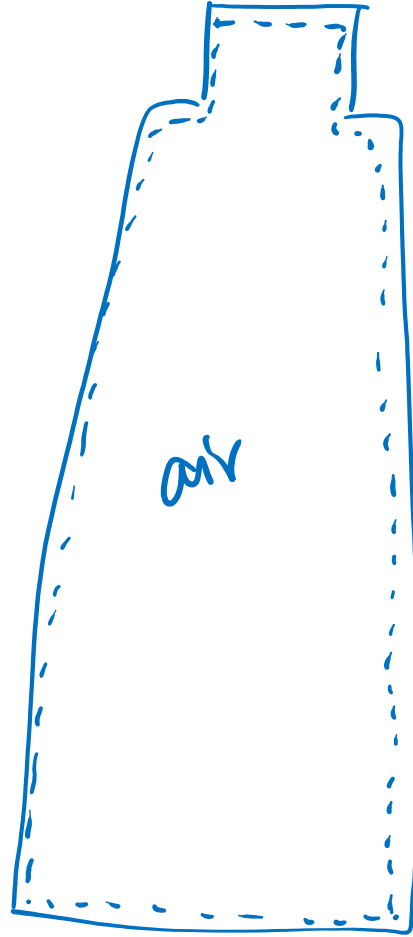
# Systems

→ fixed mass of material separated from its surroundings by a boundary

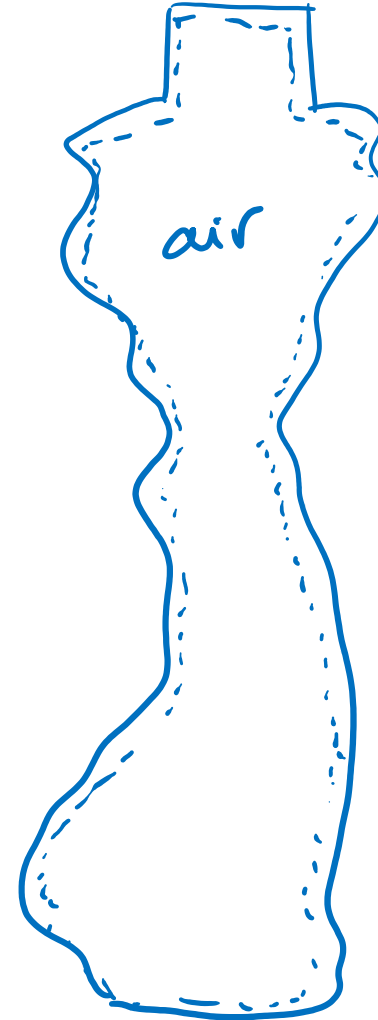


→ isolated,  $E_{in} = E_{out} = 0$

# Example: Water bottle on a plane



40,000 ft



0 ft?



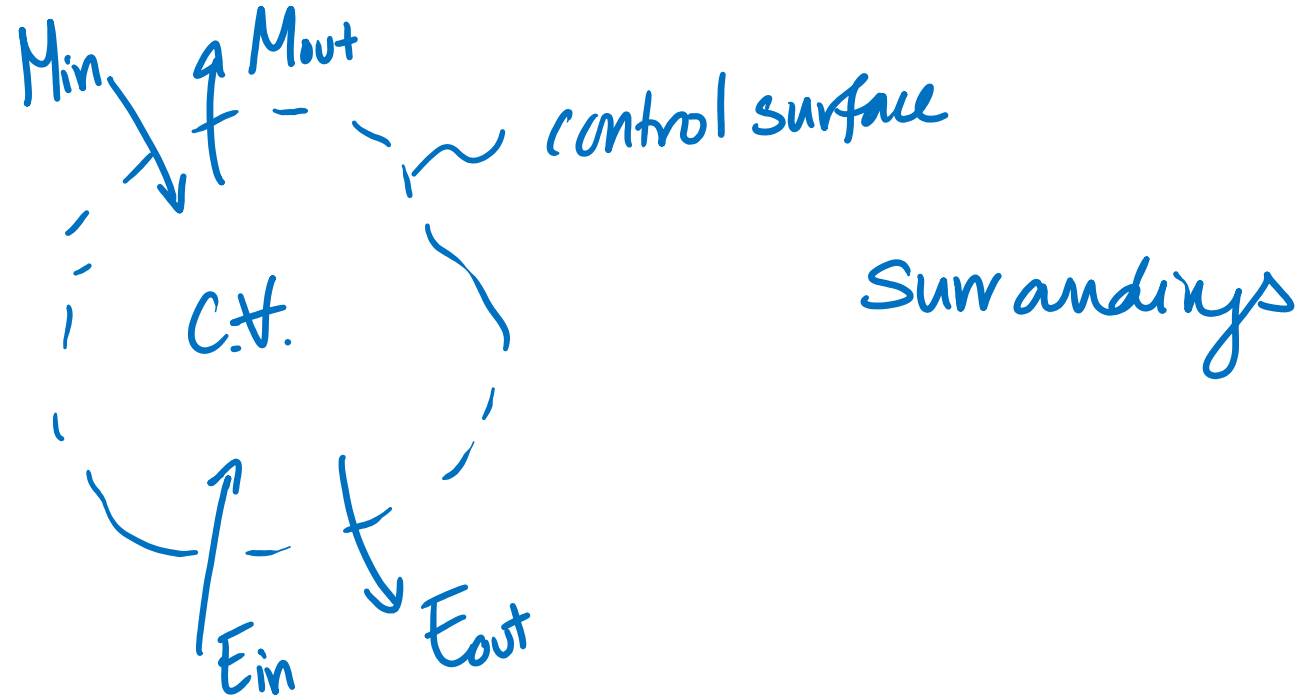


What happens to the water bottle after we land if I closed it at 40,000 feet altitude?

- a) Expands
- b) Contracts
- c) Stays the same

# Control volumes $\rightarrow$ flow

C.V.: region of space surrounded by a control surface, through which both mass and energy can pass



# Example: Fire hose







Should we use a system or control volume to analyze flow from a fire hose?

- a) System
- b) Control volume

Example: Fire hose  $\rightarrow$  flow  $\rightarrow$  control volume

